

Computational Microelectronics

Lecture 12

Sung-Min Hong (smhong@gist.ac.kr)

Semiconductor Device Simulation Laboratory

Department of Electrical Engineering and Computer Science

Gwangju Institute of Science and Technology (GIST)

L12

Issues

- We have written a code to calculate I_D .
 - ~~1) Electron continuity equation is analytically solved.~~
 - ~~2) MOS equation and continuity equation are separately treated.~~
 - ~~3) Mobility is assumed to be constant. (Partially covered)~~
 - 4) Short-channel MOSFET
 - 5) Source/drain regions
 - 6) Transient effect
 - 7) Multiple devices (Can we calculate the VTC of a CMOS inverter?)

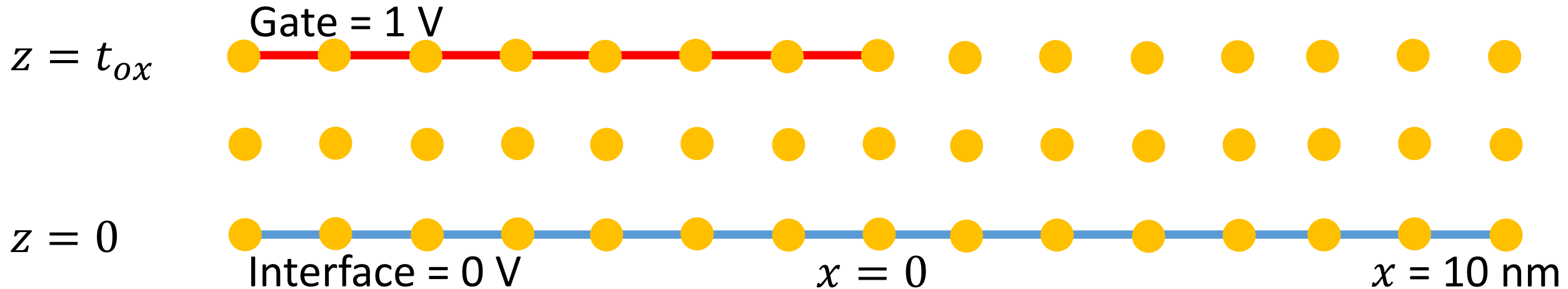
Two-dimensional Laplace equation

- Inside the oxide, there is no charge.

- Then, the equation becomes

$$\nabla^2 \phi = \frac{\partial^2}{\partial x^2} \phi + \frac{\partial^2}{\partial z^2} \phi = 0$$

- Solve it for this simple structure.



Discretization of $\nabla^2 \phi$ in 1D

- It is just $\frac{d^2 \phi}{dx^2}$.

– For uniform spacing of Δx , at $x = x_i$,

$$\left. \frac{d^2 \phi}{dx^2} \right|_{x_i} = \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{(\Delta x)^2}$$

– Integration from $x_{i-0.5}$ to $x_{i+0.5}$

$$\begin{aligned} \left. \frac{d^2 \phi}{dx^2} \right|_{x_i} &\approx \frac{1}{x_{i+0.5} - x_{i-0.5}} \int_{x_{i-0.5}}^{x_{i+0.5}} \frac{d}{dx} \left[\frac{d\phi}{dx} \right] dx \\ &= \frac{1}{x_{i+0.5} - x_{i-0.5}} \left[\left. \frac{d\phi}{dx} \right|_{x_{i+0.5}} - \left. \frac{d\phi}{dx} \right|_{x_{i-0.5}} \right] \end{aligned}$$

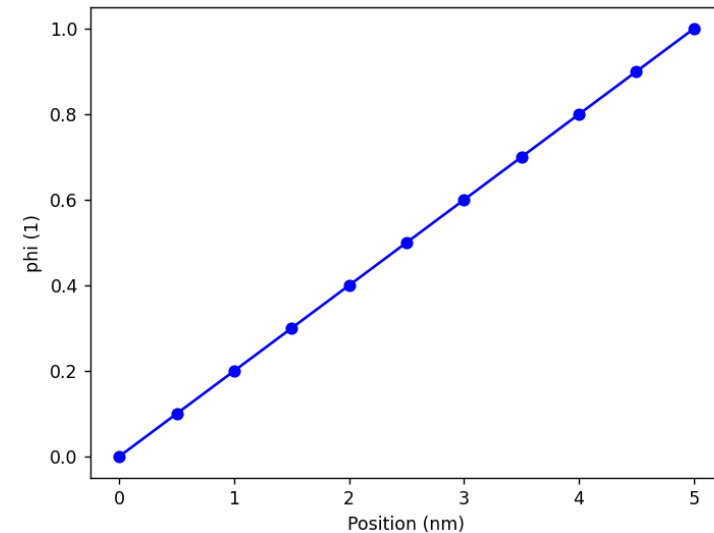
Laplace equation in 1D

- The Laplace equation, $\frac{d^2\phi}{dx^2} = 0$ (in its integrated form)

– At $x = x_i$,

$$\frac{\phi_{i+1} - \phi_i}{x_{i+1} - x_i} - \frac{\phi_i - \phi_{i-1}}{x_i - x_{i-1}} = 0$$

- Of course, we need the boundary condition.
 - Specify ϕ at the leftmost and rightmost points.



Divergence theorem

- Volume integral of $\nabla \cdot \mathbf{F}$ over a domain Ω ($\mathbf{F}$ is a vector field.)

$$\int_{\Omega} \nabla \cdot \mathbf{F} d^3r$$

- Surface integral of the normal component of $\mathbf{F}$ over the surface of Ω , $\partial\Omega$ ($\mathbf{n}$ is the outward unit normal vector.)

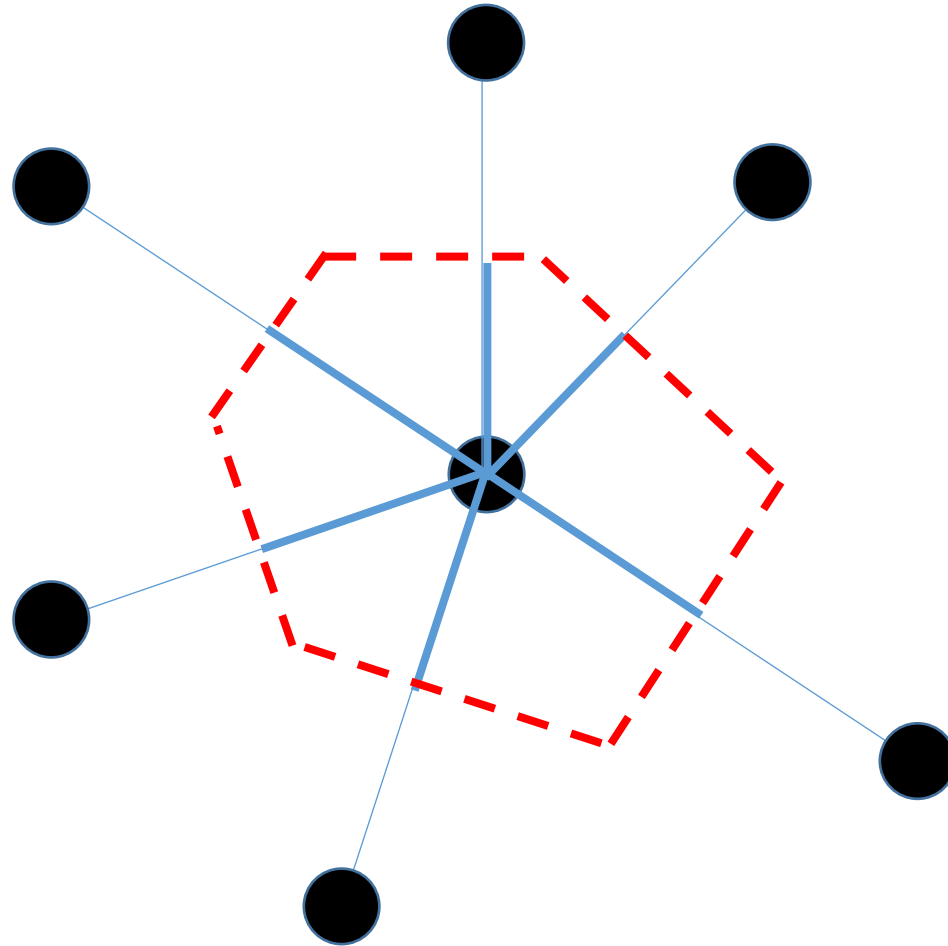
$$\int_{\partial\Omega} \mathbf{F} \cdot \mathbf{n} da$$

- Divergence theorem states that

$$\int_{\Omega} \nabla \cdot \mathbf{F} d^3r = \int_{\partial\Omega} \mathbf{F} \cdot \mathbf{n} da$$

Control volume, Ω_i

- Set of points whose nearest vertex is $\mathbf{r}_i$



In 2D or 3D,

- Integration runs over Ω_i .

– Then, following the divergence theorem,

$$\int_{\Omega_i} \nabla^2 \phi \, d^3 r = \int_{\partial \Omega_i} \nabla \phi \cdot \mathbf{n} \, da$$

– At $\mathbf{r} = \mathbf{r}_i$,

$$\int_{\partial \Omega_i} \nabla \phi \cdot \mathbf{n} \, da \approx \sum_j \nabla \phi \cdot \mathbf{n}_{ji} A_{ij}$$

Area (in 3D)

Neighboring vertices

Unit vector from $\mathbf{r}_i$ to $\mathbf{r}_j$

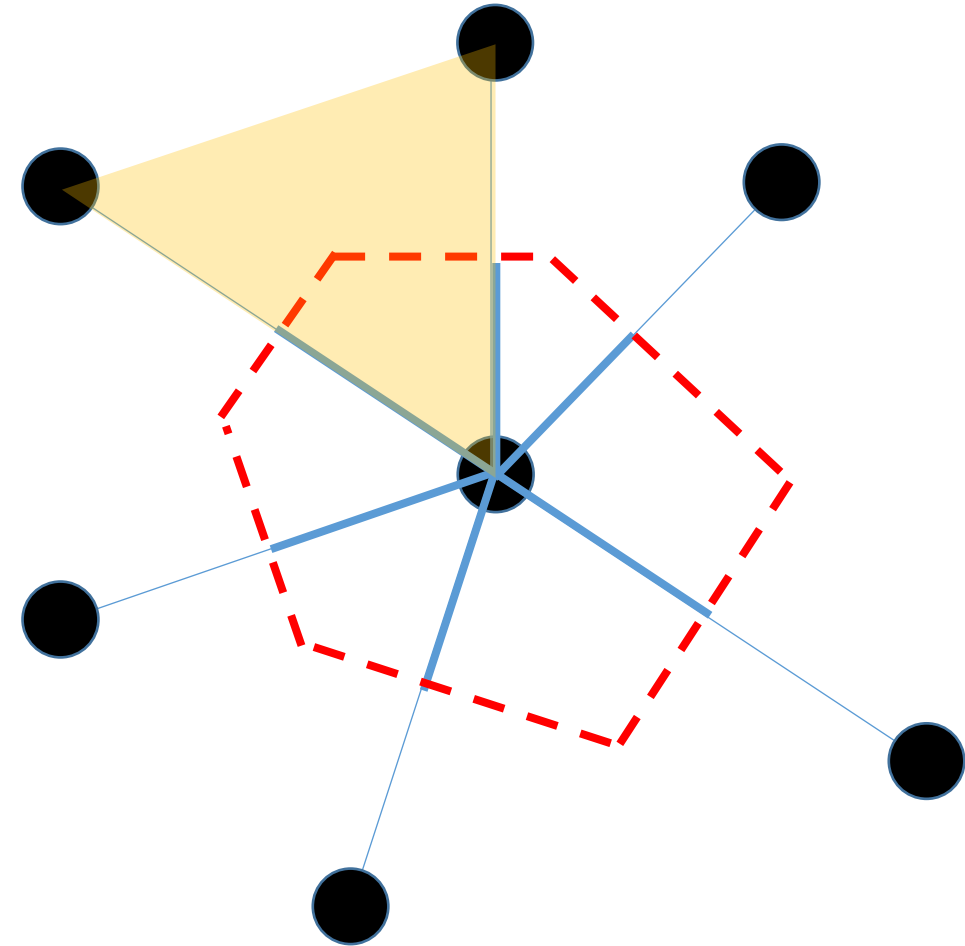
Directional approximation

- We need to evaluate $\nabla\phi \cdot \mathbf{n}_{ji}$.
 - It is well approximated as $\frac{\phi_j - \phi_i}{|\mathbf{r}_j - \mathbf{r}_i|}$.
 - Therefore, the (integrated and discretized) Laplacian operator at $\mathbf{r} = \mathbf{r}_i$ reads

$$\int_{\Omega_i} \nabla^2 \phi \, d^3r \approx \sum_j \frac{\phi_j - \phi_i}{|\mathbf{r}_j - \mathbf{r}_i|} A_{ij}$$

How can we calculate A_{ij} ?

- The distance, $|\mathbf{r}_j - \mathbf{r}_i|$, is easy to calculate.
 - Then, we must calculate A_{ij} .
 - In 2D structures, find the circumcenter of each triangle.



Laplace equation in 2D or 3D

- The Laplace equation, $\nabla^2 \phi = 0$ (in its integrated form)

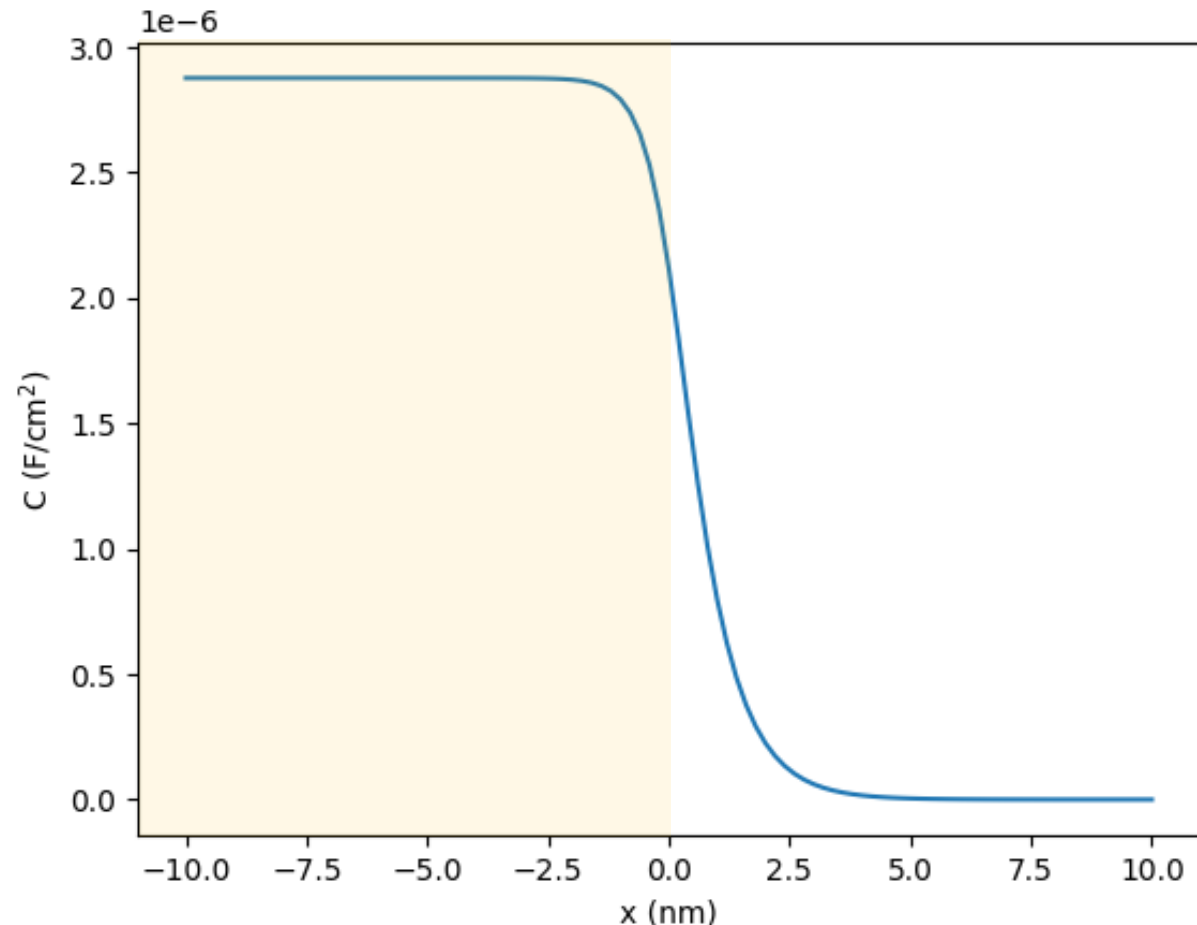
- At $\mathbf{r} = \mathbf{r}_i$,

$$\sum_j \frac{\phi_j - \phi_i}{|\mathbf{r}_j - \mathbf{r}_i|} A_{ij} = 0$$

- Of course, we need the boundary condition.
 - Specify ϕ at the contact points. (Dirichlet boundary condition)
 - For other boundary points, leave them without modification. (Neumann boundary condition)

1.2-nm-thick oxide

- A structured mesh is adopted.
 - The capacitance is calculated as
$$C(x) = \epsilon_{ox} \frac{|E_z(x, 0)|}{V_G}$$
 - The capacitance decreases rapidly as x increases.



Thank you!